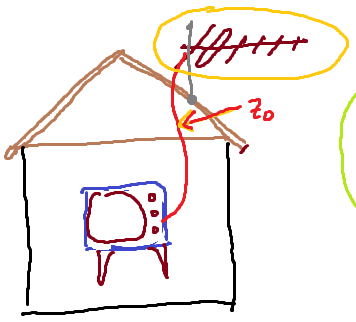
Asep Barnas Simanjuntak ①



Defenisi

- Webster : " a usually metalic device (such as a rod or wire) for radiating or receiving radio waves. "
- IEEE : " a means for radiating or receiving radio waves. " ✓

$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{20}{120} = \frac{1}{6} \rightarrow VSWR = \frac{1+1/6}{1-1/6} = \frac{1.167}{0,833} = 1,4$ ✓ → 1.5 → R/L = ? ④

$R/L = -20 \log|1/6| = 15,56\ dB = 16\ dB = 40 \rightarrow \frac{1}{40}$ →

VSWR = 2

R/L = 10 dB → $10\ dB = 10^{\frac{10}{10}} = 10$

$= 10 \rightarrow \frac{1}{10}$

RL = 20 dB = 100

R/L = 13 dB = $10^{\frac{13}{10}} = 10^{1.3} = 20$

R/L = 17 dB = 56

RL = 13 dB = 20 → artinya ada daya pantul sebesar $\frac{1}{20}$

$\rightarrow \frac{1}{20} = |\Gamma|^2 \rightarrow |\Gamma| = 0,2236$

$\rightarrow VSWR = \frac{1+0,2236}{1-0,2236} = 1.576$

2.4 GHz Wi-Fi

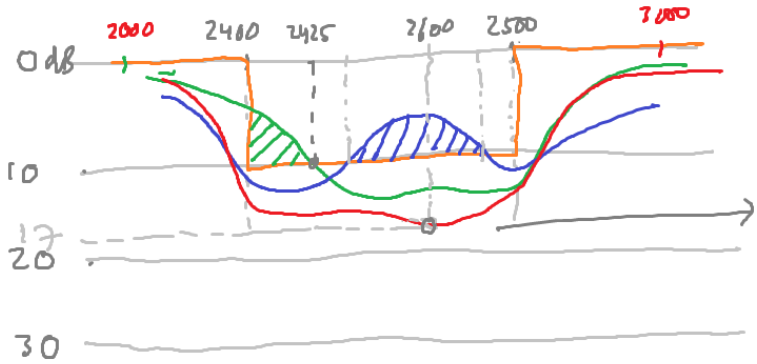
2500 MHz
2400 – 2486 MHz → R/L = 10 dB

VSWR = 2 → 10%

VSWR = 3 → ?

R/L = 13.5 dB = $10^{1.35}$ = 22,39 = 22 → $\frac{1}{22}$ → $\frac{10}{22}$ = 0,45W

→ 9.55 W



17 dB = $10^{1.7}$ = 50

$\frac{1}{50}$

Wi-Fi ⇒ 2.4GHz

2500 MHz
2400 – 2486 MHz → R/L = 10 dB

